Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

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The RDME: reactions and diffusion across a spatial grid

Model. Divide a 1-D domain into K voxels. Each voxel k holds molecule counts $\mathbf{n}_k \in \mathbb{Z}^S_{\geq 0}$.

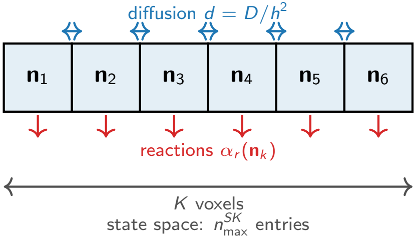
Two types of events:

- **Reactions** inside voxel k: rate $\alpha_r(\mathbf{n}_k)$, change $\mathbf{n}_k$.
- **Diffusion** between neighbors: rate $d = D/h^2$, moves one molecule $k \leftrightarrow k+1$.

The joint probability $p(\mathbf{n}_1, \ldots, \mathbf{n}_K, t)$ satisfies $\dot{p} = Ap$ **(RDME)**.

The computational bottleneck

The state space has $\sim n_{\max}^{SK}$ reachable states — **exponential in** K. Even K=8, S=1 is intractable with a naive FSP.



$\dot{p} = Ap$ **has block-tridiagonal structure** (schematic, K=4, S=1)

$$\begin{array}{cc} & \begin{array}{cccc} \mathbf{n}_1 & \mathbf{n}_2 & \mathbf{n}_3 & \mathbf{n}_4 \end{array} \\ \begin{array}{c} \mathbf{n}_1 \\ \mathbf{n}_2 \\ \mathbf{n}_3 \\ \mathbf{n}_4 \end{array} & \begin{bmatrix} A_1 & D_{12} & & \\ D_{12} & A_2 & D_{23} & \\ & D_{23} & A_3 & D_{34} \\ & & D_{34} & A_4 \end{bmatrix} \end{array} \cdot \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix} = \begin{bmatrix} \dot{p}_1 \\ \dot{p}_2 \\ \dot{p}_3 \\ \dot{p}_4 \end{bmatrix}$$

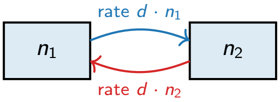
each block: $n_{\max}^S \times n_{\max}^S$ total: $n_{\max}^{SK} \times n_{\max}^{SK}$

A_k: reactions inside voxel k (dense block) $D_{k,k+1}$: diffusion between neighboring voxels (sparse: shifts one count)

The Binomial is the exact diffusion equilibrium

Fix total molecules $\bar{n} = n_1 + n_2$ across two adjacent voxels.

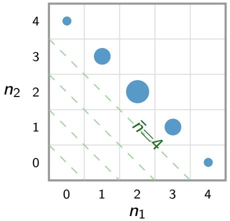
Diffusion moves one molecule at a time, at rate d per molecule:



Each molecule hops independently $\Rightarrow$ at equilibrium, equally likely to be in either voxel. For $\bar{n}$ total:

$$\boxed{P(n_1 \mid \bar{n}) = \mathrm{Bin}\big(\bar{n}, \tfrac{1}{2}\big)} \quad (\text{exact, any } d > 0)$$

When the distribution is near this equilibrium, $\bar{n}$ **alone describes the two-voxel state**.



Probability spreads along constant-total diagonals as $\mathrm{Bin}(\bar{n}, \frac{1}{2})$ (dot size $\propto$ mass).

The two-voxel state is captured by $\bar{n}$ alone.

Merging two adjacent voxels: restriction and prolongation

Compress ($\mathcal{R}$): exact:

$$p^c(\bar{n}) = \sum_{n_1+n_2=\bar{n}} p(n_1, n_2).$$

Reconstruct ($\mathcal{P}$): approximate:

$$p(n_1, n_2) \approx p^c(\bar{n}) \cdot \pi(\bar{n}),$$

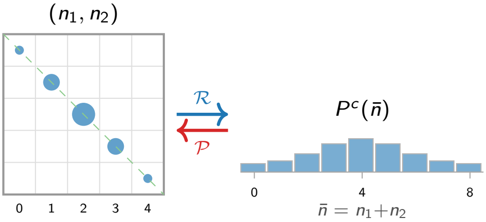
where π is the **within-pair distribution**.

State-space reduction

Per pair: $n_{\max}^{2S} \to n_{\max}^{S}$.
All K voxels: $n_{\max}^{SK} \to n_{\max}^{SK/2}$.

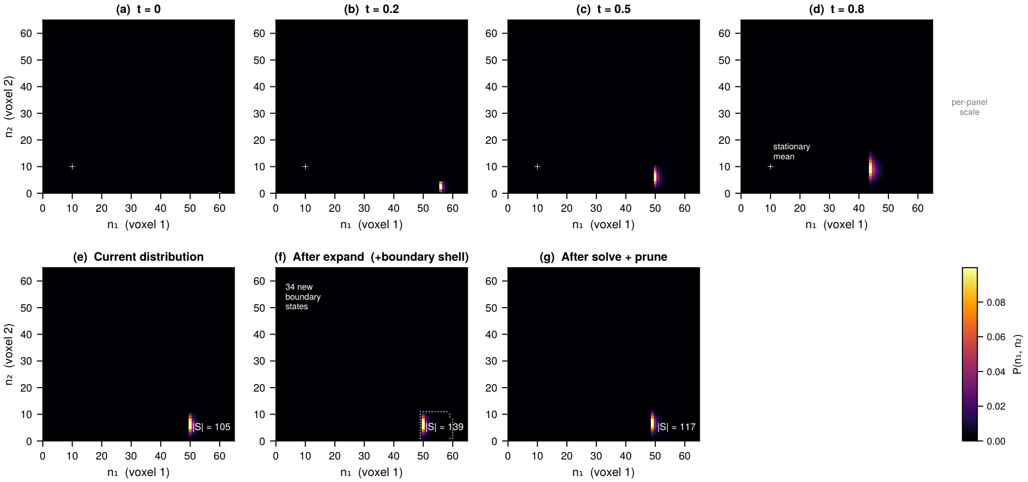
Classical choice: $\pi = \mathrm{Bin}(\bar{n}, \frac{1}{2})$ — exact when diffusion $\gg$ reactions.

New: patch QSD (later) — exact conditional



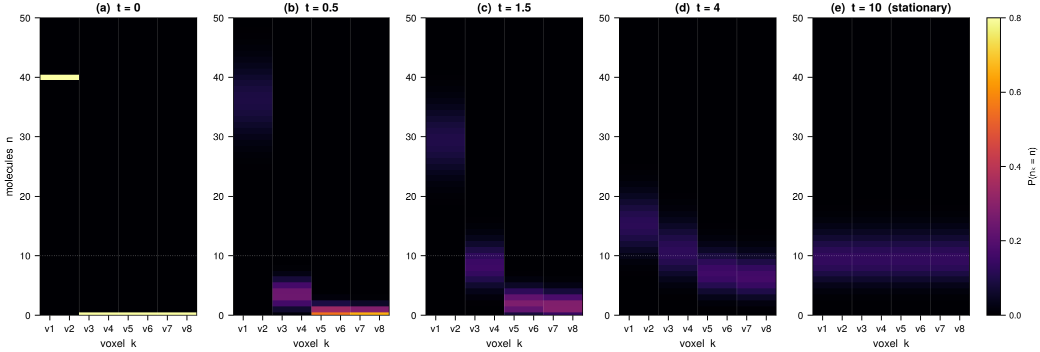
$\mathcal{R}$: sum along each diagonal $\to$ scalar $\bar{n} = n_1+n_2$.
$\mathcal{P}$: spread back via within-pair distribution π.

The adaptive FSP algorithm: expand, solve, prune



Top row: joint distribution $P(n_1, n_2)$ of a 2-voxel birth-death system: point source at $(n_{\mathrm{src}}, 0)$ traveling to the stationary point (white cross).

Result: K=8 RDME wavefront via coarsened simulation



K=8 wavefront solved via K=4 coarse FSP. Max coarse state space: 106585 states. K=8 direct FSP would be intractable (grows to 90k+ states by t=0.8). Fine marginals recovered analytically via Binom($n\hat{c}_j$, ½) prolongation — exact for balanced pairs.

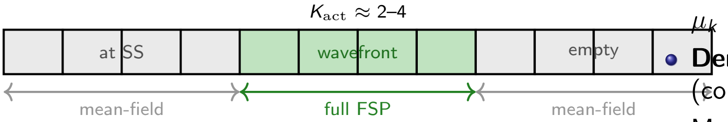
Birth-death RDME, $K=8$, $D=0.1$. IC: $v_1=v_2=40$ (delta), $v_3 \ldots v_8=0$. Each panel shows marginals $P(n_k=n)$ for all 8 voxels.

(a)–(d): Wavefront sweeps left→right, distributions broaden. **(e):** Stationary: all voxels reach Poisson(10) (dashed line).

Solved via **K=4 coarse FSP** (30k states). Direct K=8 FSP: intractable. Fine marginals recovered via Bin$(\tilde{n}, \frac{1}{2})$:

Mixed resolution: only the wavefront needs fine tracking

Key observation: at any time t, the probability distribution has *three* spatial zones:



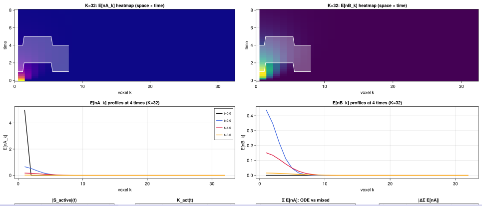
- **At steady state** (behind front): $P(\mathbf{n}_k)$ is near-Poisson$(\mu_{\mathrm{ss}}) \Rightarrow$ mean-field μ_k is sufficient.
- **Empty** (ahead of front): $P(\mathbf{n}_k) \approx \delta_0 \Rightarrow$ mean-field $\mu_k \approx 0$.
- **Wavefront**: full joint FSP on K_{act} voxels. $K_{\mathrm{act}} \ll K$, *independent of* K.

State-space bound

$|S_{\mathrm{active}}| = O\left(n_{\max}^{2SK_{\mathrm{act}}}\right)$ regardless of K.

Adaptive active set:

- **Promote** voxel to FSP when $\mu_k > \theta_{\mathrm{promote}}$ (wavefront arriving).
- **Demote** to mean-field when $\mu_k \approx \mu_{\mathrm{ss}}$ (converged) or $\mu_k \approx 0$ (empty).
- Mean-field ODE for inactive voxels — *exact for linear reactions.*
- Hard cap $K_{\mathrm{act}}^{\max}$ prevents blowup for fast diffusion.

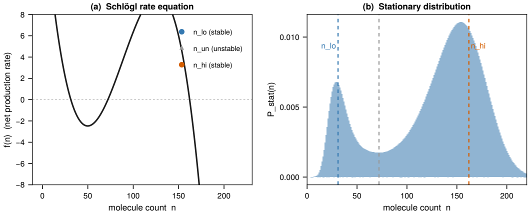


Bistable reactions

For birth-death, each voxel has one stable state.
Diffusion wins: the pair is always well-mixed.
Binomial coarsening is *essentially exact*.

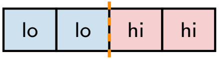
But what if reactions are bistable?

The **Schlögl model** has two stable states per voxel:
$n_{\text{lo}} \approx 31$ and $n_{\text{hi}} \approx 162$.



Spatial metastability

Neighbors can occupy *different* states. A **spatial front** forms.



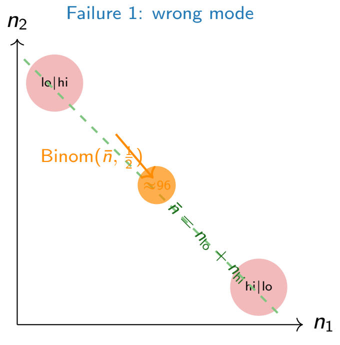
Reactions "pull" voxels apart; diffusion "pulls" them together. When reactions are fast, the front **persists for very long times**.

Failure mode 1: bistable front (Schlögl)

The joint distribution $P(n_1, n_2)$ is **bimodal**: mass at $(n_{\mathrm{lo}}, n_{\mathrm{hi}})$ and $(n_{\mathrm{hi}}, n_{\mathrm{lo}})$.

Restriction loses orientation

$\mathcal{R}(n_{\mathrm{lo}}, n_{\mathrm{hi}}) = n_{\mathrm{lo}} + n_{\mathrm{hi}} = \mathcal{R}(n_{\mathrm{hi}}, n_{\mathrm{lo}})$: **the front direction is erased**. $\mathrm{Binom}(\bar{n}, \frac{1}{2})$ places mass at the **unstable** midpoint.



Failure mode 2: bimolecular front (A+B$\to \emptyset$)

Binomial assumes n_A and n_B are **independent** within a pair. But A+B$\to \emptyset$ drives them **apart**: a molecule of A and one of B quickly annihilate if they share a sub-voxel.

For strong k_{rxn}, Binomial **overestimates** P(co-located) by $\sim$19$\times$, inflating the within-patch reaction rate.

k_{rxn}	$\rho(n_A, n_B)$	P(co-located)
0	0.00	76.6% *(Binom)*
5	−0.80	36.4%
50	−0.98	**4.0%**

The patch QSD: physics-informed prolongation

One unified fix for both failure modes.

Replace Binomial with the **quasi-stationary distribution (QSD)** of the local patch dynamics:

$$\pi(n_1, \ldots, n_M \mid \bar{\mathbf{n}}) = \text{QSD of } M\text{-voxel local CME}$$

where the local CME includes internal diffusion *and* reactions within the patch, with reactions that leave the $\bar{\mathbf{n}}$ subspace treated as **absorbing** (QSD conditioning).

Generality

M=2 (1D pair) · **M=4** (2D 2×2) · **M=8** (3D 2×2×2).
Works for any `CoarseningMap` topology.
Precomputed once: one small eigenproblem per $(\bar{n}_A, \bar{n}_B)$.

QSD vs Binomial at the bimolecular front

(d=2, $\bar{n}_A$=$\bar{n}_B$=3, varying k_{rxn}):

k_{rxn}	$\rho(n_{A,1}, n_{B,1})$	$P(\mathrm{co-located})$
0	$+0.000$	76.6%
0.5	-0.095	75.7%
5	-0.796	36.4%
50	-0.982	**4.0%**

Binomial: always 76.6% (independent assumption).

QSD corrects the $\sim$19× overestimate at strong k_{rxn} — *without any additional runtime cost* beyond the precomputed table.

For covered states, the **multiplicative update** $p^{\mathrm{new}}(x) = \tilde{p}(x) \cdot \bar{p}^{\mathrm{post}} / \bar{p}^{\mathrm{pre}}$ preserves the fine-level

The V-cycle with conditional prolongation

One V-cycle step:

1. **Pre-smooth**: intra-pair diffusion for τ_{pre}
2. **Restrict**: $\bar{n}_j = n_{2j-1} + n_{2j}$
3. **Recurse**: V-cycle on $K/2$ voxels
4. **Prolong**: conditional reconstruction
5. **Prune**: remove states with $p(x) < \tau$

Prolongation strategy:
Covered states ($\bar{x} \in \mathcal{R}\tilde{p}$): multiplicative update

$$p^{\mathrm{new}}(x) = \tilde{p}(x) \cdot \frac{\bar{p}^{\mathrm{post}}(\bar{x})}{\bar{p}^{\mathrm{pre}}(\bar{x})}$$

Asymmetry-preserving: mass at $(n_{\mathrm{lo}}, n_{\mathrm{hi}})$ stays there.
New states (expand!-added): use **patch QSD** $\pi(\bar{n})$ — the within-patch conditional (details

Why not reconstruct at every step?

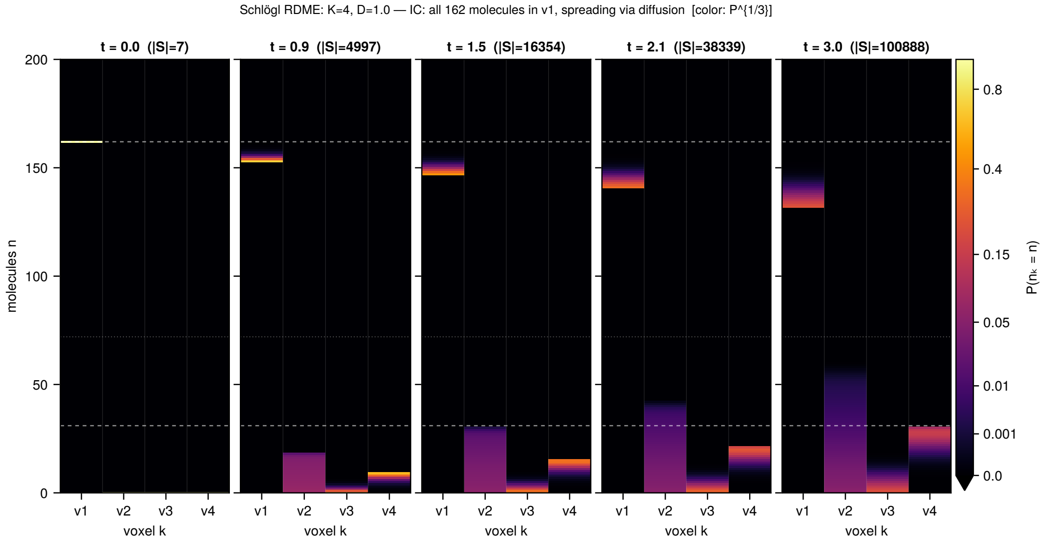
Forcing a Binomial split on every pair, every step, erases the within-pair asymmetry (the bistable front).

Instead: evolve p^h directly. Coarsening is *only* for the recursive correction; the fine window tracks the true distribution exactly.

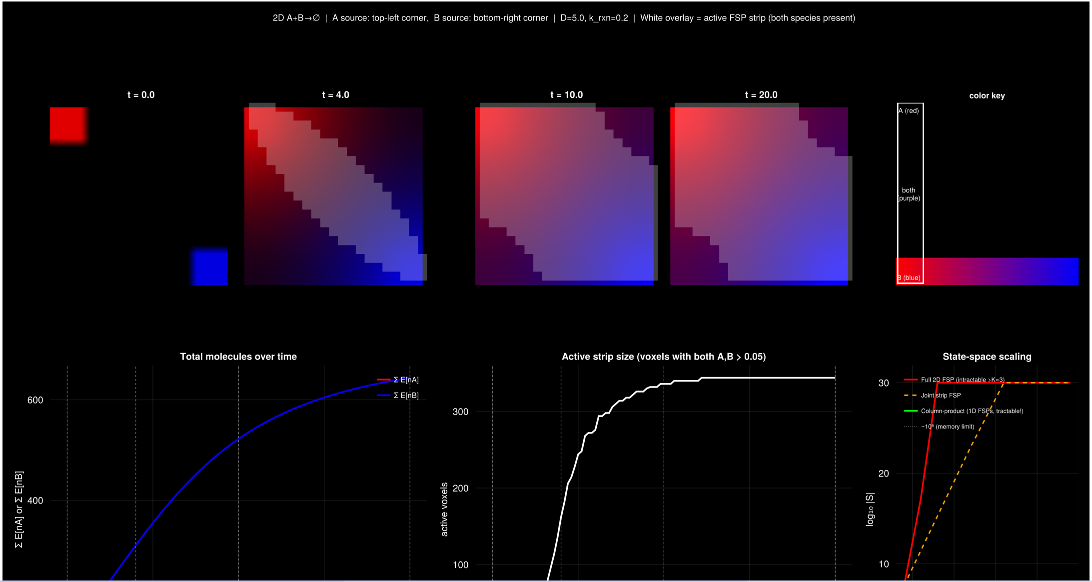
Diffusion boundary states ($L_1{=}2$)

Coarse diffusion $(\bar{n}_1, \bar{n}_2) \to (\bar{n}_1 - 1, \bar{n}_2 + 1)$ is always $L_1{=}2$. Without this neighbor search, **inter-pair mass transport is invisible** to the V-cycle and molecules never leave pair 1.

Result: Schlögl wavefront spreading via the V-cycle



Two-species A+B→ ∅: mixed-resolution reaction front



Summary

1. Restriction + prolongation compresses the RDME state space.

When the within-patch distribution is near-Binomial/Multinomial, marginalizing patches reduces the state space from $n_{\max}^{2S}$ to $n_{\max}^{S}$ per patch. *K=8 birth-death wavefront solved with 30k states (direct: intractable).*

2. Conditional prolongation handles bistable and bimolecular fronts.

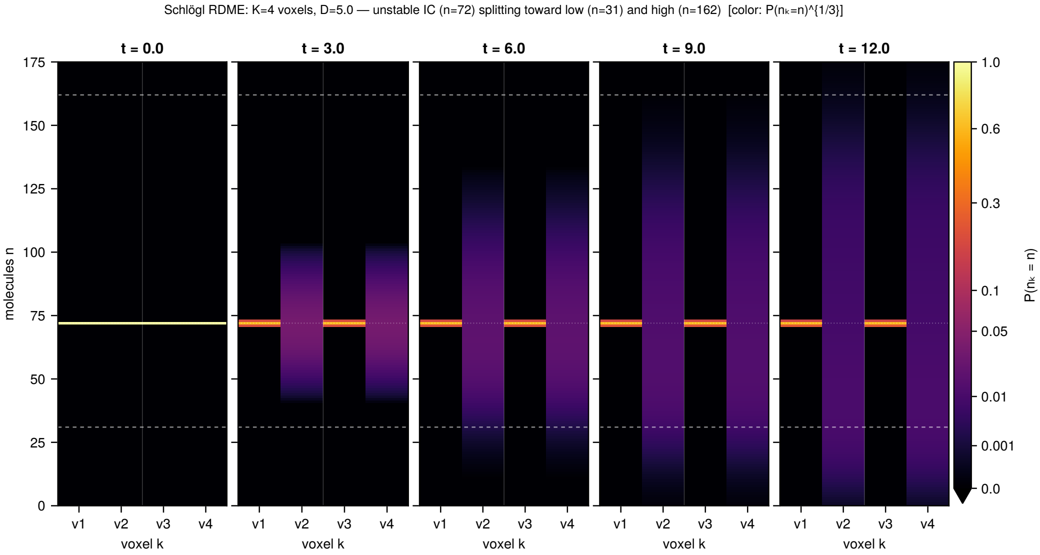
Multiplicative update for covered states preserves within-pair asymmetry. $L_1 \leq 2$ neighbor search propagates both reaction and diffusion boundaries. *Schlögl wavefront and bistable splitting correctly captured.*

3. Mixed-resolution active set: the wavefront is always narrow.

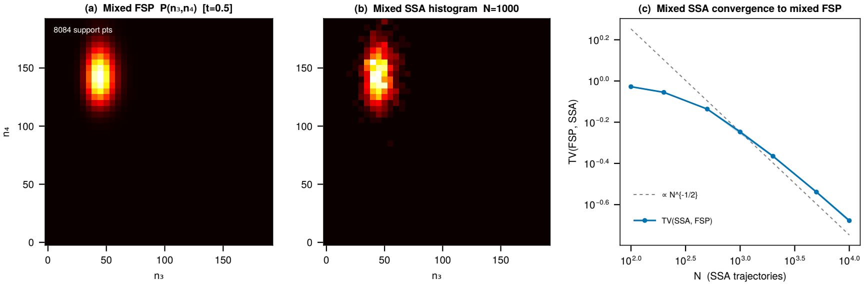
Mean-field for inactive voxels (exact for linear reactions; QSD closure for bimolecular). Adaptive promote/demote keeps $K_{\mathrm{act}} = O(1)$ regardless of K. *$A+B \to \emptyset$ reaction front tracked in 2D; column-product gives linear scaling in K_y.*

4. Patch QSD: physics-informed prolongation for bimolecular reactions.

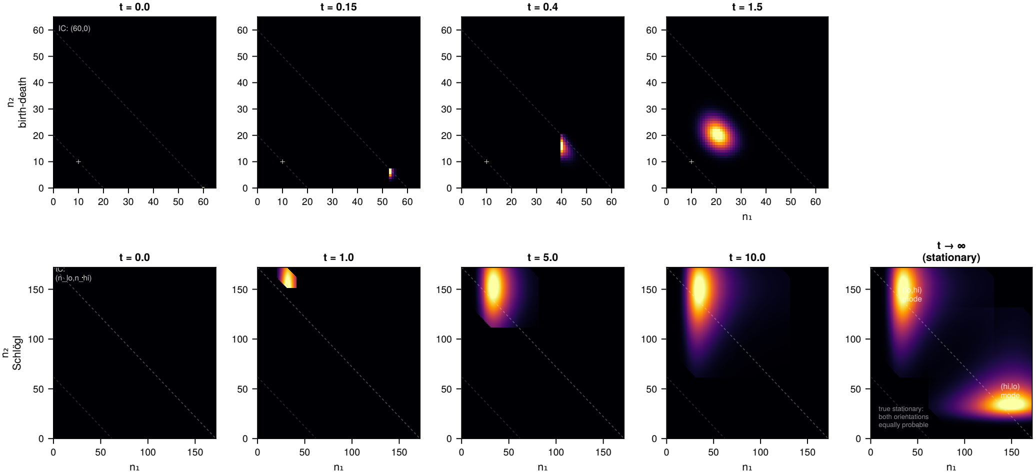
Appendix: Schlögl splitting from the unstable fixed point



Appendix: mixed-resolution generator self-consistency



ODE solve on the mixed state $(\bar{n}_1, n_3, n_4)$ vs. Monte Carlo on the same reduced generator. Means agree to $<1\%$; TV $\sim N^{-1/2}$ (sampling noise only).



Top: birth-death point source traveling to Poisson(10) stationary. Binomial coarsening exact.

Bottom: Schlögl front IC persisting metastably; true stationary (rightmost) shows both orientations equally.

Appendix: K=8 persistent front